



# KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

Telephone: 080 -23600978, 23341652 || Email: spp@kscst.org.in  
Website: www.kscst.iisc.ernet.in/spp.html or www.kscst.org.in/spp.html

## FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 46<sup>th</sup> SERIES OF STUDENT PROJECT PROGRAMME

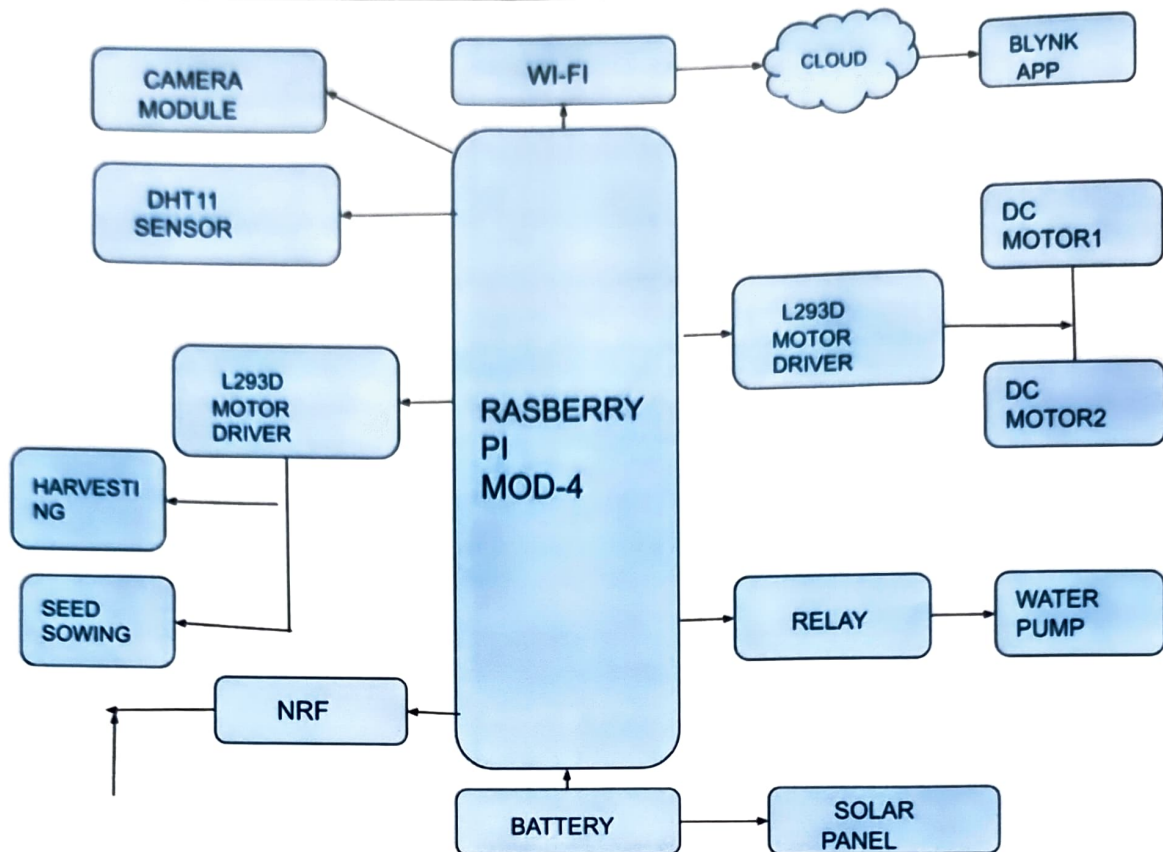
(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form. Send the softcopy of the project proposal including the three scanned pages and send the proposal (All information in one pdf file) by email to spp@kscst.org.in

<https://forms.gle/pMfzw4iKL7LNAojd8>

|    |                                                                                                                                                                                                                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <b>Name of the College: PRESIDENCY UNIVERSITY</b>                                                                                                                                                                                                                                                                                                                     |
| 2. | <b>Project Title: SWARM ROBOTICS FOR IRRIGATION (SRI)</b>                                                                                                                                                                                                                                                                                                             |
| 3. | <b>Branch: ELECTRONICS AND COMMUNICATION ENGINEERING</b>                                                                                                                                                                                                                                                                                                              |
| 4. | <b>Theme (as per KSCST poster):</b><br><b>(The project proposals shall mandatorily be from one of the broad themes / areas. Visit website <a href="http://www.kscst.org.in/spp.html">www.kscst.org.in/spp.html</a>)</b><br><br><b>AUTOMATION OR NEW CONCEPTS IN AGRICULTURE</b>                                                                                       |
| 5. | <b>Name(s) of project guide(s):</b><br>1. <b>Name: Prof. / Dr. M.S.DIVYA RANI</b><br><b>Email id: divyarani@presidencyuniversity.in</b><br><b>Contact No.: 9480313352</b><br>2. <b>Name: Prof. / Dr. Mohammed Mujahid Ulla Faiz</b><br><b>Email id: mohammed.mujahid@presidencyuniversity.in</b><br><b>Contact No.: 7406994135</b>                                    |
| 6. | <b>Name of Team Members (Strictly not more than four students in a batch):</b> (Type names in Capital Letters as provided in your college)<br>(Please paste the latest passport size photograph adjacent to your respective names)<br><b>Name: KANDULA JITHENDRA SUBRAMANYAM</b><br><b>USN No.:20191ECE0148</b><br><b>Email id: jithendrasubramanyam218@gmail.com</b> |

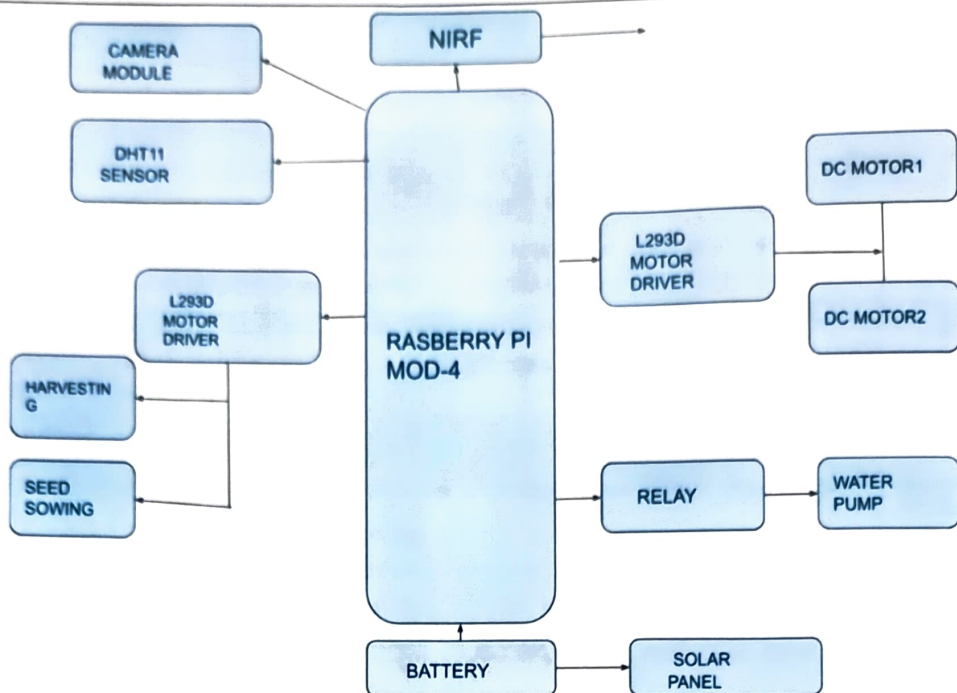
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>Mobile No:9704400336</b><br/> <b>Name: KANTHETI TARUN MANIKANTA</b><br/> <b>USN No.:20191ECE0152</b><br/> <b>Email id: tarunmanikanta007@gmail.com</b><br/> <b>Mobile No.: 9866141499</b></p> <p><b>Name: KATTA RAMPRASAD</b><br/> <b>USN No.: 20191ECE0159</b><br/> <b>Email id: ramkatta23@gmail.com</b><br/> <b>Mobile No.:9494443911</b></p> <p><b>Name: KESANA KASEE VISWANADH</b><br/> <b>USN No.: 20191ECE0161</b><br/> <b>Email id: kaseeviswanadhkesanakesana@gmail.com</b><br/> <b>Mobile No.: 7386475489</b></p>                                                                                                           |
| 7.  | <p><b>Team Leader of the Project:</b><br/> <b>Name: KANDULA JITHENDRA SUBRAMANYAM</b><br/> <b>USN No.:20191ECE0148</b><br/> <b>Email id: jithendrasubramanyam218@gmail.com</b><br/> <b>Mobile No.: 9704400336</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.  | <p><b>Processing Fee Details (Through Online Payment only):</b><br/>         (processing fee of Rs. 1000/-)<br/> <b>Please furnish the payment made details provided in the last page of this proposal.</b><br/> <b>NEFT</b></p> <p><b>Note:</b> (The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the project proposal and 2) make the payment made details for processing fees and 3) Enter the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).</p>                                                                             |
| 9.  | <p><b>Date of commencement of the Project: 5/03/2023</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 10. | <p><b>Probable date of completion of the project: 10/05/2023</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11. | <p><b>Scope / Objectives of the project:</b><br/>         In India, 70% of people pick agriculture as their primary line of work. Our present generation lacks enough skilled workers, particularly in the agriculture sector. Farming by hand takes more time and produces more pollutants. The primary goal of automation in agriculture is to reduce labour and the amount of time needed to process crops, resulting in a 90 percent reduction in human effort. In the agricultural sector, robotics is anticipated to play a significant role. Multi-robot systems and collaborative strategies are frequently suggested as viable</p> |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>remedies to increase system robustness and efficiency. Swarm robotics, which emphasises flexibility, scalability, and robustness in completing complicated tasks, is one of the multi-robot techniques and is thought to be particularly useful for precision farming and large-scale agricultural applications. This Agribot offers a cutting- edge way for harvesting and watering the crops with the least amount of labour and labour, making it a useful machine. Also, an Android smartphone can be used to control the vehicle online. Motors and sensors that are interfaced with embedded computers are built for the entire calculation, processing, and monitoring process. The robot is managed using Internet of Things technology.</p> <ul style="list-style-type: none"> <li>• The main aim of this Project is to Propose an IOT based Agribot and agriculture data acquisition using swarm robotics.</li> <li>• To measure temperature and humidity in the field and turn on the Water pump.</li> <li>• To harvest the field and weed cutting.</li> <li>• To sow the seeds</li> </ul> <p>Pest detection and control.</p> |
| 12. | <b>Methodology:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



**BLOCK DIAGRAM OF MASTER ROBOT**





BLOCK DIAGRAM OF SLAVE ROBOT

Swarm robotics is the coordination of a large number of multirobots with straightforward physical robots. This behaviour reveals how a robot interacts with its surroundings and other robots. Swarm activity can be seen in a variety of settings, including artificial swarm intelligence, biological studies of insects and ants, and other natural settings. A fresh and creative project idea was generated using the survey literatures and after consulting several technical papers. The goal of the SRI project is to create a team of robots that can develop a schematic input provided by the user on a certain platform. By attaching numerous sensors and actuators to the main board of the Raspberry Pi platform, IoT networks may be created, allowing robots to interact with one another and complete their duties on their own. In our project, there are two robots: a master robot and a slave robot. Swarm robotics dictates that the slave robot will follow the master robot's instructions. In our project work, we will be developing a self learning systems which makes the decisions on the entire irrigation process through sensing technology. Initial instructions will be given to the Master robot via Mobile App. Various features we will be working is as follows.

## MECHANISM

### • Ploughing

The first and foremost step in the farming is ploughing. This process is done in order to loosen the soil and create a path or tracks on the farm land in order to sow the seeds uniformly. The structure and the design of the plough

tool depends on the various constraints such as the type of soil to be ploughed and the depth required based on the type of crop that has to be grown and so on. There are many types of ploughing mechanisms that have been adopted which can be broadly classified into two categories; one is the manually driven ploughing tool and the other being machine driven. The design and dimensions of the plough tool are in accordance with the size of the bots. The angle of inclination and length of the tool are calibrated by considering the depth required for ploughing the soil and it varies with the type of crops and soil. The tool is operated by using a 12V dc servomotor. The initial and final positions of the plough tool are controlled by coding it in a required manner.

#### • **Seeding**

The next major step in the process of farming is seeding. Seeding usually depends on the type of crops being grown and the type of seeding varies over a variety of crops. In case of robots, utmost care has to be taken to ensure uniform spacing and controlled flow of the seeds from the bot; where in the seeds required for sowing is stored in a container and it is mounted on the bot at the suitable position.

#### • **Irrigation/ Fertilize**

Based on the amount of water required for that particular crop suitable irrigation methods are adopted. Our bot inculcates the drip irrigation for the irrigation process. The water required for the irrigation is stored in a container and is mounted onto the bot. The frequency of drop of the water is controlled using water pump. The water from the container flows to the field through the structure provided for irrigation uniformly. The same irrigation tool can be used for fertilizing by replacing the water with fertilizer. The mechanism can also be further automated by directly sending the commands from the bots to the main water pump for irrigation.

#### • **Harvesting**

The final process under farming is harvesting where in the crops are cut down or chopped using the designed harvesting tool. The harvesting tool is driven by the dc motor which rotates the harvesting tool at a rate of 1500rpm which is sufficient enough to cut down the crops in order to separate the grains from the crop. The present bot only performs the crop chopping action and doesn't involve any process regarding the grain separation from the crop.

#### • **Pest Detection and Control.**

The system is equipped with Pi -cam to analyse the plants growth. An efficient machine learning algorithm called CNN is developed with SVM classifiers to recognize the plant disease. A suitable pesticides will be sprayed with the suitable communication to monitoring station.

Hence the application of swarm intelligence principles to agricultural robotics

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>can lead to disruptive innovation. The project work aims at demonstrating such advantages and comparing them with the current state of the art within the context of a monitoring/mapping scenario</p> <p><b>Note:</b> In case of fabrication work in the project, an engineering drawing with dimensions / detailed design should be attached to the proposal.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13. | <p><b>Expected Outcome of the project:</b></p> <p>The three major criteria that drive our system are (1) Speed, (2) Accuracy, (3) Flexibility and (4) Low Cost.</p> <p>The systems mentioned above are either product specific or can handle equipment in only. Diversity in delivery systems has yet to be achieved and the system proposed in this paper completely satisfies all required multitasking capabilities and bridges the gap between the warehouse and the industrial activities.</p> <p>Overall, the use of swarm robotics in agriculture has the potential to improve efficiency, reduce costs, and improve crop yields, while also minimizing environmental impact. However, more research is needed to fully understand the potential outcomes of swarm robotics in agriculture and to develop effective and practical swarm robot systems for use on farms.</p> <ul style="list-style-type: none"> <li>• It drastically reduces the manpower thereby eliminating human error, thereby achieving maximum efficiency.</li> <li>• Each robotic unit in the proposed system costs around \$100 to \$120. Therefore a complete system can be easily incorporated in a few thousands of dollars.</li> <li>• It uses simple control software to control hundreds of autonomous mobile robots; this robotic fulfilment system enables extremely fast cycle times, from receiving to picking to shipping.</li> <li>• The system is completely modular, which means you can start with a small system, and add more gear as your business grows.</li> <li>• The size of the robots isn't one of the constraints as it should possess enough mechanical power for heavy lifting.</li> </ul> <p>Overall, the use of swarm robotics in agriculture has the potential to improve efficiency, reduce costs, and improve crop yields, while also minimizing environmental impact. However, more research is needed to fully understand the potential outcomes of swarm robotics in agriculture and to develop effective and practical swarm robot systems for use on farms.</p> |



14. **Is the project proposed relevant to the Industry / Society or Institution?**

**Yes / No:**

**If Yes, please provide details of the Industry / institution and contact details:**

**Yes both industry and institution**  
**Presidency University Launch Pad**

**(Note:** Preference will be given to those projects relevant to the industry / institution. Hence be specific in giving detailed information). Is the industry extending support - technology / funds / use the final product, please specify.

15. **Can the product or process developed in the project be taken up for filing a Patent?**

**Yes / No: Yes**

**Prior Art search done?**

**Yes/No: No**

**Note:** If your answer is "Yes", you may contact Patent Information Centre of KSCST. For more details, email: [pic@kscst.org.in](mailto:pic@kscst.org.in)

16. **Budget details (break-up details should be given):**

**Note:** KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee.

| <b>Budget</b>                               | <b>Amount</b> |
|---------------------------------------------|---------------|
| a) Materials / Consumables (Please specify) |               |
| raspberry pi mod4 *2                        | 17000         |
| zigbee module *2                            | 4000          |
| camera module *2                            | 900           |
| DC motors *8                                | 800           |
| servo motors *5                             | 850           |
| relay module *2                             | 300           |
| rechargeable batteries *6                   | 600           |
| b) Labor (Describe)                         | 0             |
| c) Travel (Describe)                        | 500           |



|     |                                                                                                                                                                                                                                                                                                                                                                        |         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|     | e) Miscellaneous (Please specify)                                                                                                                                                                                                                                                                                                                                      | 1000    |
|     | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                           | 25950/- |
| 17. | <b>Any other technical details (Please specify):</b><br>NIL                                                                                                                                                                                                                                                                                                            |         |
| 18. | <b>SPP Coordinator (Identified by the college):</b><br><b>Note:</b> To be identified by the principal of the institution. The project proposals must be submitted to KSCST through SPP coordinator designated by the Principal.<br><br><b>Name: Dr.RAMESH C S Dean R&amp;D</b><br><b>Email id: deanresearch@presidencyuniversity</b><br><b>Contact No.: 9880904891</b> |         |

**Name of the Project Guide: Dr M.S Divya Rani**  
**Email id: [divyarani@presidencyuniversity.in](mailto:divyarani@presidencyuniversity.in)**  
**Contact No.: 9480313352**

**Name of the HOD: Dr. Rajiv Ranjan Singh**  
**Email id: [hod\\_ece@presidencyuniversity.in](mailto:hod_ece@presidencyuniversity.in)**  
**Contact No.: 9742649493**

## **DECLARATION**

### **(From Project Students)**

(To scan this page and enclose in the project proposal)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: ...SWARM ROBOTICS FOR IRRIGATION..., Branch: ...ELECTRONICS AND COMMUNICATION ENGINEERING..., College: ...PRESIDENCY UNIVERSITY.....) are true and correct to the best of our knowledge and belief and we undertake to inform KSCST of any changes therein in the project title, students name will be intimated immediately through project guide. In case any of the above information is found to be false or untrue or misleading, we are aware that we may be held liable for it. We hereby authorize sharing of the project information with this project proposal with the Karnataka State Council for Science and Technology, Bengaluru.

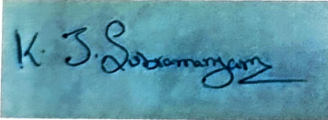
We are aware that the project team must exhibit / demonstrate the project in the nodal centre and interact regarding project with the experts and to exhibit the project in the State Level Seminar and Exhibition (if selected). If the student team fails to attend the evaluation in nodal centre or fails to attend the State Level Seminar and Exhibition, the supported project amount will be returned to KSCST.

We also hereby, enclose the endorsement form to KSCST, Bengaluru.

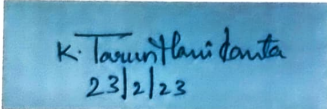
**Name of the students with USN No.**

**Signature with date**

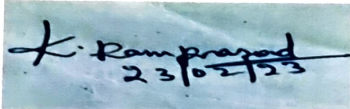
1. Kandula Jithendra Subramanyam 20191ECE0148

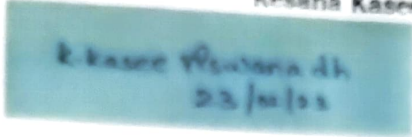


2. Kantheti Tarun Manikanta 20191ECE0152



3. Katta Ramprasad 20191ECE0159





(Name &amp; Signature of Project Guide with Seal)

Email id: divyarani@presidencyuniversity.in

Contact No.: 9480313352

(Name &amp; Signature of HOD with Seal)

Email id: hod\_ece@presidencyuniversity.in

Contact No.: 9742649493

## ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that 1) Mr.KANDULA JITHENDRA SUBRAMANYAM, 2) Mr. KANTHETI TARUN MANIKANTA 3) Mr. KATTA RAMPRASAD..., 4) Mr. KESANA KASEE VISWANADH ,are bonafide student(s) of Department of ELECTRONICS AND COMMUNICATION ENGINEERING , in the degree program of our institution. If the project proposal submitted by these students under the 46<sup>th</sup> series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the nodal centre and in the State Level Seminar and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in nodal centre or fails to attend the State Level Seminar and Exhibition, the supported project amount will be returned to KSCST.

(Name &amp; Signature of Project Guide with Seal)

Email

id:divyarani@presidencyuniversity.in

Contact No.:9480313352

(Signature of HOD with Seal)

School of Engineering  
PRESIDENCY UNIVERSITY  
Rajalakshmi, Yelahanka, Bengaluru -64

Email

id:hod\_ece@presidencyuniversity.in

Contact No.:9742649493

(Signature of the Principal with Seal)

Email

id:pro-vicechancellor@presidencyuniversity.in

Contact No.:7348892241

Dr. A.M. SURENDRA KUMAR  
Pro Vice Chancellor  
Presidency University, Bengaluru



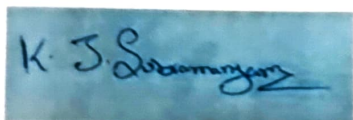
## DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(Note: Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

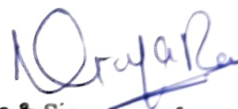
|                            |   |                                   |
|----------------------------|---|-----------------------------------|
| 1. TITLE OF THE PROJECT    | : | SWARM ROBOTICS IRRIGATION (SRI)   |
| 2. NAME OF THE TEAM LEADER | : | KANDULA JITHENDRA SUBRAMANYAM     |
| 3. EMAIL ID                | : | jithendrasubramanyam218@gmail.com |
| 4. CONTACT MOBILE NO.      | : | 9704400336                        |

### PAYMENT MADE DETAILS

|                                                           |   |                                              |
|-----------------------------------------------------------|---|----------------------------------------------|
| 5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)          | : | CIUBH23053002482                             |
| 6. TRANSACTION ID                                         | : |                                              |
| 7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER | : | KANTHETI VENKATA SRINIVASA RAO<br>9949494448 |
| 8. NAME OF THE BANK                                       | : | CITY UNION BANK                              |
| 9. PROCESSING FEES                                        | : | Rs. 1000/-                                   |
| 10. DATE OF PAYMENT MADE                                  | : | 22/02/2023                                   |
| 11. TIME                                                  | : | 11AM                                         |
| 12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)     | : | NEFT                                         |



(Name & Signature of  
the team leader)



(Name & Signature of  
Project Guide or HOD with Seal)

# KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

## 46<sup>th</sup> SERIES OF STUDENT PROJECT PROGRAMME (SPP)

*(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)*

### BANK ACCOUNT DETAILS OF KSCST

|                                     |                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------|
| Name and address of the Institution | Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012 |
| Account holder's name / Designation | Secretary, Karnataka State Council for Science and Technology                      |
| Bank Account No. & Name of the bank | Current A/C No. 0683201000024<br>Canara Bank, IISc Campus Branch, Bangalore-560012 |
| IFSC Code                           | CNRB0000683                                                                        |
| MICR Code                           | 560015023                                                                          |
| Bank Branch Address                 | Canara Bank, Indian Institute of Science, Bangalore-560012                         |

### BANK DETAILS

|                                        |                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------------|
| Name of the Agency                     | Karnataka State Council for Science and Technology<br>IISc Campus, Bangalore - 560012 |
| Account holder's name / Designation    | Secretary , Karnataka State Council for Science and Technology                        |
| Bank Account No. &<br>Name of the bank | Current A/C No. 0683201000024<br>Canara Bank IISc Campus Branch<br>Bangalore-560012   |
| IFSC Code                              | CNRB0000683                                                                           |
| MICR Code                              | 560015023                                                                             |
| Bank Branch Address                    | Canara Bank<br>Indian Institute of Science<br>Bangalore-560012                        |